

Quantum chemical studies on a family of Ketocoumarin photosensitizers

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Contents

1	Introduction and Motivation	2
2	Molecules Studied in this Work	2
3	Computational Details	3
4	Results	4
4.1	Conformational sampling and DFT optimizations	4
4.2	UV-Vis absorption spectra	7
4.3	Potential-Energy Curves along Important Torsions	9
4.3.1	The Rotation of the Amine Group	10
4.3.2	The Rotation of the Phenoxy Group	12
5	Conclusions & Outlook	15
A	Brief Theoretical Background	17
A.1	Electronic Schrödinger Equation and the Born–Oppenheimer Approximation	17
A.2	Kohn–Sham Density-Functional Theory	17
A.3	Exchange–Correlation Functionals and TPSSh	17
A.4	Basis-Set Representation	18
A.5	Excited States and TD-DFT	18

1 Introduction and Motivation

The conversion of solar energy into chemical energy is an important objective in research and especially based on the modern goals of humanity. Light-driven hydrogen evolution represents one possible strategy for storing solar energy in the form of a chemical fuel. In these photocatalytic systems, a photosensitizer absorbs incident light and transfers energy or an electron to a catalyst. Many commonly used photosensitizers contain scarce and expensive transition metals such as ruthenium, iridium, or platinum and this motivates the development of efficient organic photosensitizers that are free of both metals and heavy atoms [1, 2].

Coumarins are a well-established class of organic chromophores whose photophysical properties can be modified through substitution of the molecular framework. Ketocoumarins (KCs) form a subclass of coumarins characterized by an additional carbonyl group at the 3-position of the chromen-2-one framework. They can exhibit high molar extinction coefficients, efficient intersystem crossing (ISC), relatively small singlet-triplet energy gaps, and long-lived triplet excited states [1]. These properties make ketocoumarins promising metal-free photosensitizers for photochemical and photocatalytic applications.

Knorr, Peneva, and co-workers recently demonstrated the application of ketocoumarins as photosensitizers for light-driven hydrogen evolution [1]. The investigated ketocoumarins achieved turnover numbers approaching 4000 with respect to the cobaloxime catalyst. Their spectroscopic and electrochemical investigations supported a reductive quenching mechanism. Hydrogen production could be sustained for up to 22 days by increasing the cobaloxime concentration and thereby reducing photodegradation of the photosensitizer. These results demonstrate the synthetic accessibility, tunability, and potential of ketocoumarins as alternatives to heavy-metal-based photosensitizers [1].

A non-radiative, twisted intramolecular charge-transfer (TICT) state was believed to play an important role for the ISC to the T_1 state. [3, 4, 5]. However, this charge-separated TICT state leads to chemical decomposition, most likely a stepwise N-dealkylation, which was not tackled in the previous publication [6]. Several modifications of the molecular scaffold have been suggested within the CataLight consortium [7]. Moreover, the photophysical properties of ketocoumarins depend not only on their chemical composition but are also expected to depend on their molecular conformation. Rotation of the amine and phenoxy groups can change their orientation relative to the central ketocoumarin framework. These torsional motions can influence conjugation, orbital overlap, excitation energies, oscillator strengths, and the relative energies of the singlet and triplet states.

The aim of this study is to investigate the conformational and electronic properties of a series of structurally modified ketocoumarins, with quantum chemical methods. This will serve as a first step towards further studies on their decomposition and their performance as photosensitizers. The conformational space was explored using the GOAT algorithm together with the g-xTB method. Representative conformers were selected manually and subsequently optimized at the DFT level of theory. Relaxed ground-state scans were performed along the amine and phenoxy torsional coordinates. TD-DFT calculations were then performed at the resulting geometries to construct ground- and excited-state potential-energy curves (PECs) and to analyze changes in excitation energies and oscillator strengths. Further details on the calculations can be found in Section 3

2 Molecules Studied in this Work

The molecular structures investigated in this work are shown in Figure 1. Structure **A** corresponds to the ketocoumarin from the original report[1], which showed the best results in terms of photocatalysis. Structure **A^d** is the deuterated analogue of **A**. Although its chemical performance may be different than **A**, it will not be studied herein since the electronic structure will be the same. Structure **B** contains no α -hydrogens. Structure **C** features a cyclic amine substituent. Structure **D** contains a conformationally restricted amine unit with reduced rotational freedom. These structural modifications were considered since they present different solutions to the problem of decomposition. For example, the absence of α -hydrogens in **B**, prohibits one of the first steps of the dealkylation mechanism, that is the abstraction of α -hydrogens. Their different structural and photophysical properties will be investigated.

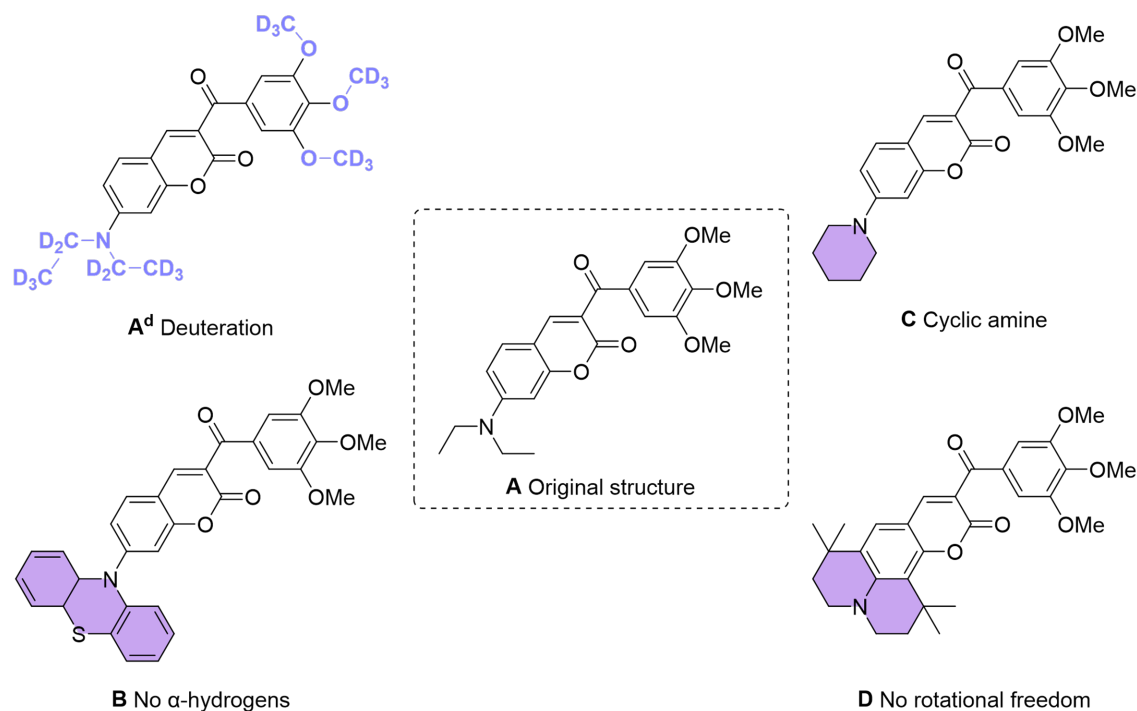


Figure 1: Molecular structures investigated in this work. The structural modifications with respect to the original KC structure are highlighted in purple.

3 Computational Details

All quantum-chemical calculations were performed using the ORCA program, version 6.1.1, for the DFT and TD-DFT calculations [8, 9]. The xTB program[10] was used for the initial geometry optimizations and conformational sampling with the g-xTB method [11]. The programs used for the visualization of molecular geometries, orbitals, and densities were GaussView6, VMD, and xyzrender [12, 13, 14]. For every ketocoumarin molecule, the following computational workflow was applied:

1. A starting molecular geometry was generated using GaussView6. An initial geometry optimization was performed using the semiempirical g-xTB method as implemented in the xTB program.
2. Conformational sampling was subsequently performed using the GOAT algorithm as implemented in ORCA [15]. The g-xTB method was used during the conformational sampling [11]. The calculations were performed using the ORCA Python Interface (OPI) [16]. An energy threshold of 7 kcal mol⁻¹ was used.
3. The conformers generated by the GOAT calculations were manually inspected. Representative conformers were selected based on the orientation of the phenoxy group and the amine group. The selected structures were further optimized at the DFT level of theory. The TPSSH[17] density functional was used together with the D4 dispersion correction[18]. The def2-TZVPP[19, 20] basis set was used for the DFT calculations. Solvation effects were described implicitly using the conductor-like polarizable continuum model[21]. A dielectric constant of 46.961 was used to approximate the experimentally employed methanol–water mixture with a 7:3 composition. Frequency calculations were performed for every optimized structure. The absence of imaginary frequencies was used to confirm that the optimized structures corresponded to minima on the ground-state potential-energy surface.
4. TD-DFT calculations were performed for every optimized conformer using the Tamm–Dancoff approximation. Sixty singlet states and sixty triplet states were calculated. Gaussian-broadened absorption

spectra were generated for the individual conformers ($\sigma = 500 \text{ cm}^{-1}$). The individual conformer spectra have been combined according to their Boltzmann populations.

5. Relaxed ground-state geometry scans were performed along the dihedral coordinate describing rotation of the amine group. Relaxed ground-state geometry scans were also performed along the dihedral coordinate describing rotation of the phenoxy group. The relaxed scans were performed using the same DFT level of theory described above. Each relaxed scan consisted of 24 geometries. TD-DFT calculations were performed at every geometry obtained from the relaxed scans. The relative ground-state energy was calculated with respect to the lowest-energy geometry of each scan. The vertical excitation energies were added to the corresponding relative ground-state energies. The resulting energies were used to construct the potential-energy curves of the singlet and triplet excited states.

The RIJCOSX approximation was used to accelerate the DFT and TD-DFT calculations. The def2/J auxiliary basis set was used. Tight SCF convergence criteria were applied using the `TightSCF` keyword. Tight geometry-optimization criteria were applied using the `TightOPT` keyword. During some relaxed geometry scans, small conformational changes of methyl groups produced significant energy differences. These conformational changes introduced discontinuities into some of the potential-energy curves. Reverse scans along the same torsional coordinates have been performed to identify geometries that produce rather continuous and physically meaningful potential-energy curves. To test whether this is due to our scanning parameters, we repeated the relaxed scan for the diethylamine-derivative by including 36 points and tight optimization criteria, and the potential energy curve was exactly the same.

4 Results

4.1 Conformational sampling and DFT optimizations

The conformational space of the four ketocoumarins was explored using the GOAT algorithm in combination with the g-xTB method. For each ketocoumarin, the conformational search generated a large number of structures within the predefined energy window of 7 kcal mol^{-1} . The choice of the energy window was done based on the fact that semi-empirical methods may give energies (and their ordering) that is not as good as DFT (although the g-xTB method seems to outperform low and mid-level density functionals) [11].

Therefore, to make sure that all important and possible conformations are taken into account we used a somewhat larger energy window of 7 kcal mol^{-1} . The number of conformers depends also on the molecular structure and flexibility of the corresponding ketocoumarin.

The conformers produced by the GOAT-g-xTB search were ordered according to their g-xTB energies. The conformer with index 0 corresponded to the lowest-energy structure, while the conformer energy generally increased with increasing conformer index. For each ketocoumarin, the lowest-energy conformer was examined first. The remaining conformers were then inspected in order of increasing energy to identify geometrically distinct arrangements of the amine and phenoxy groups. Inspection was done manually by visualizing the geometries in `VMD` and/or `GaussView6`. The results of the conformational sampling are summarized in Figure 2, where we overlay the conformer geometries.

For the diethylamine-substituted ketocoumarin, four representative conformers were selected. After alignment with respect to the ketocoumarin core, the conformers could be grouped into four main geometrical arrangements. These arrangements were distinguished primarily by the orientation of the phenoxy group, which could occupy four different spatial quadrants with respect to the approximately planar coumarin core. The four quadrants arise from the placement of the phenoxy group on either side of the coumarin plane ('up/down') and toward either side of the molecular framework ('left/right'). Smaller conformational differences were also observed within each group, mainly due to rotations of the ethyl substituents of the diethylamine fragment. For each of the four main phenoxy-group arrangements, the lowest-energy conformer belonging to that arrangement was selected.

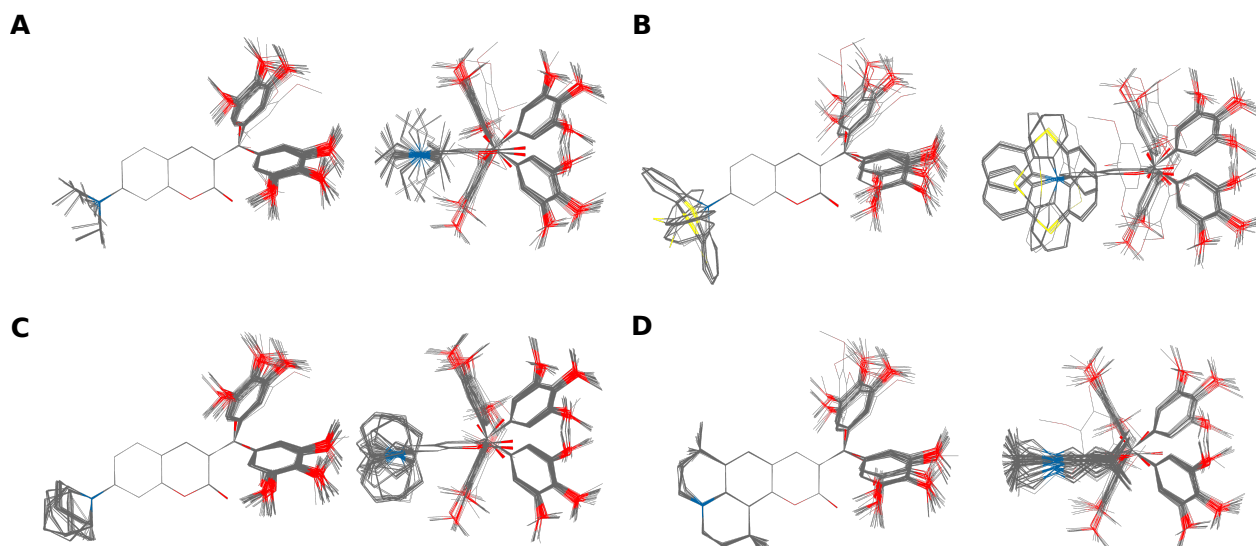


Figure 2: The conformational space of ketocoumarins **A–D**. For each ketocoumarin, all conformer geometries obtained from the conformational sampling are overlaid and shown from two different perspectives.

For the phenothiazine-substituted ketocoumarin, ten representative conformers were selected because both the phenoxy group and the phenothiazine-containing fragment showed several distinct orientations. Four conformers were selected to represent the four characteristic spatial quadrants occupied by the phenoxy group relative to the approximately planar ketocoumarin core. The remaining six conformers were selected to represent different orientations of the phenothiazine-containing fragment with respect to the ketocoumarin plane. These structures included arrangements in which the phenothiazine fragment was approximately planar, tilted, or nearly perpendicular to the ketocoumarin core. For each identified geometrical arrangement, the lowest-energy conformer belonging to that arrangement was selected.

For the piperidine-substituted ketocoumarin, four representative conformers were selected. Just like the diethylamine analogue, the principal distinction between these conformers was the placement of the phenoxy group in four different spatial quadrants with respect to the approximately planar ketocoumarin core. Smaller structural differences were also observed in the piperidine fragment, including variations in the orientation of its chair-shaped ring relative to the ketocoumarin framework. For each of the four main phenoxy-group arrangements, the lowest-energy conformer identified during the visual inspection was selected.

For the quinolizine-substituted ketocoumarin, four representative conformers were selected. The fused quinolizine fragment is conformationally restricted and remains in a similar orientation relative to the ketocoumarin core across the conformer ensemble. The main distinction between the selected conformers was therefore the orientation of the phenoxy group, which occupied four different spatial quadrants with respect to the approximately planar ketocoumarin core. The lowest-energy conformer was retained together with three additional low-energy conformers representing the remaining phenoxy-group arrangements.

To summarize, the manual selection procedure therefore yielded four representative conformers for the diethylamine derivative, ten for the phenothiazine derivative, four for the piperidine derivative, and four for the quinolizine derivative. These structures were used as starting geometries for the subsequent DFT optimizations.

The selected conformers were subsequently optimized using ground-state DFT at the TPSSh-D4/def2-TZVPP level of theory. We chose to optimize only a handful of conformers for each molecule to reduce the computational time and because the differences between conformers that belong to the same cluster are small and concern only different orientations of methyl groups. This higher-level treatment was used to refine the geometries obtained from the lower-cost g-xTB conformational search. The good performance of the g-xTB method was proven from the quick optimization convergence of our DFT calculations. Frequency calculations were performed at the same level of theory for each optimized conformer. No imaginary fre-

quencies were obtained for the optimized conformers, confirming that they corresponded to minima on the ground-state potential-energy surface.

The frequency calculations also provided us with the Gibbs free-energy corrections required for properly comparing the thermodynamic stability of the conformers. The Gibbs free energy of conformer i was obtained as

$$G_i = E_{\text{el},i} + G_{\text{corr},i}, \quad (1)$$

where $E_{\text{el},i}$ is the electronic energy and $G_{\text{corr},i}$ is the corresponding Gibbs free-energy correction. For each ketocoumarin, the conformer with the lowest Gibbs free energy was used as the reference (G_{min}), and the relative Gibbs free energy was calculated according to

$$\Delta G_i = G_i - G_{\text{min}}. \quad (2)$$

The Boltzmann population of conformer i was calculated according to

$$p_i = \frac{\exp(-\Delta G_i/RT)}{\sum_j \exp(-\Delta G_j/RT)}, \quad (3)$$

where p_i is the population of conformer i , R is the gas constant, T is the temperature, and the summation runs over all DFT-optimized conformers of the corresponding ketocoumarin. A temperature of 298.15 K was used for the calculation of the conformer populations. The relative Gibbs free energies were expressed in kcal mol^{-1} , and the corresponding value of the gas constant was used in the Boltzmann expression. The percentage population was obtained as $100 \cdot p_i$. The resulting relative Gibbs free energies and Boltzmann populations are summarized in Table 1.

Table 1: Relative Gibbs free energies and Boltzmann populations of the DFT-optimized conformers. The relative Gibbs free energies are reported with respect to the lowest-free-energy conformer of each ketocoumarin. The Boltzmann populations were calculated at 298.15 K.

Ketocoumarin	Conformer	ΔG (kcal mol ⁻¹)	Population (%)
A	conf_000	2.4115	1.63
	conf_002	0.0000	95.18
	conf_032	2.5419	1.30
	conf_042	2.3225	1.89
B	conf_000	2.4975	1.36
	conf_001	2.5309	1.28
	conf_004	0.0000	91.78
	conf_008	2.3931	1.62
	conf_018	2.4972	1.36
	conf_020	5.3181	0.01
	conf_021	2.4669	1.43
	conf_029	7.8798	0.00
	conf_031	2.5910	1.16
	conf_034	5.2017	0.01
C	conf_000	0.2089	21.52
	conf_003	0.2203	21.11
	conf_055	0.0000	30.62
	conf_056	0.0800	26.75
D	conf_000	0.0000	37.56
	conf_002	2.5812	0.48
	conf_035	0.0930	32.10
	conf_042	0.1358	29.86

For the diethylamine-substituted ketocoumarin, conf_002 was the lowest-free-energy conformer and accounted for 95.18% of the calculated equilibrium population. The remaining three conformers were ap-

proximately 2.32–2.54 kcal mol^{−1} higher in Gibbs free energy and each contributed less than 2%. The conformational distribution of the diethylamine-substituted ketocoumarin was therefore strongly dominated by this single conformer in our selection.

A similar conformational distribution was obtained for the phenothiazine-substituted ketocoumarin. Conformer *conf_004* was the lowest-free-energy structure and contributed 91.78% of the calculated equilibrium population. Six of the remaining conformers were approximately 2.39–2.59 kcal mol^{−1} above the minimum and individually contributed approximately 1–2%. The three conformers located more than 5 kcal mol^{−1} above the minimum made negligible contributions to the equilibrium population.

In contrast, the piperidine-substituted ketocoumarin exhibited a considerably broader conformational distribution. Conformer *conf_055* was the lowest-free-energy structure and contributed 30.62% of the population. Conformers *conf_056*, *conf_000*, and *conf_003* contributed 26.75%, 21.52%, and 21.11%, respectively. The relative Gibbs free-energy differences among these four conformers were smaller than 0.23 kcal mol^{−1}, indicating that all four conformers were thermally accessible at 298.15 K.

The quinolizine-substituted ketocoumarin also showed several substantially populated conformers. Conformer *conf_000* was the lowest-free-energy structure and contributed 37.56% of the population. Conformers *conf_035* and *conf_042* were only 0.0930 and 0.1358 kcal mol^{−1} above the minimum and contributed 32.10% and 29.86%, respectively. In contrast, *conf_002* was 2.5812 kcal mol^{−1} above the minimum and contributed only 0.48%.

The DFT results therefore indicate two different types of conformational behaviour within the investigated series. The diethylamine- and phenothiazine-substituted ketocoumarins were each dominated by one conformer. In contrast, several conformers contributed significantly to the equilibrium populations of the piperidine- and quinolizine-substituted ketocoumarins. These Boltzmann populations were subsequently used to weight the individual conformer contributions to the calculated UV-Vis absorption spectra.

4.2 UV-Vis absorption spectra

TD-DFT calculations within the Tamm–Dancoff approximation were performed for every DFT-optimized conformer of the four ketocoumarins. The calculated vertical excitation energies and oscillator strengths of the singlet excited states were used to construct the absorption spectrum of each conformer. The discrete electronic transitions were represented using Gaussian broadening. Each conformer spectrum was weighted according to the corresponding Boltzmann population obtained from the ground-state DFT and frequency calculations. The weighted conformer spectra were subsequently summed to obtain the final calculated UV-Vis absorption spectrum of each ketocoumarin.

The Gaussian-broadened spectrum of conformer *i* was constructed according to

$$I_i(E) = \sum_n f_{i,n} \exp \left[-\frac{(E - E_{i,n})^2}{2\sigma^2} \right], \quad (4)$$

where $E_{i,n}$ is the vertical excitation energy of transition *n* for conformer *i*, $f_{i,n}$ is the corresponding oscillator strength, and σ is the Gaussian broadening parameter (equal to 500 cm^{−1} in our case).

The final absorption spectrum of each ketocoumarin was calculated as the Boltzmann-weighted sum of the individual conformer spectra:

$$I_{\text{final}}(E) = \sum_i p_i I_i(E), \quad (5)$$

where p_i is the Boltzmann population of conformer *i* at 298.15 K. The calculated spectra of the four ketocoumarins are shown in Figure 3.

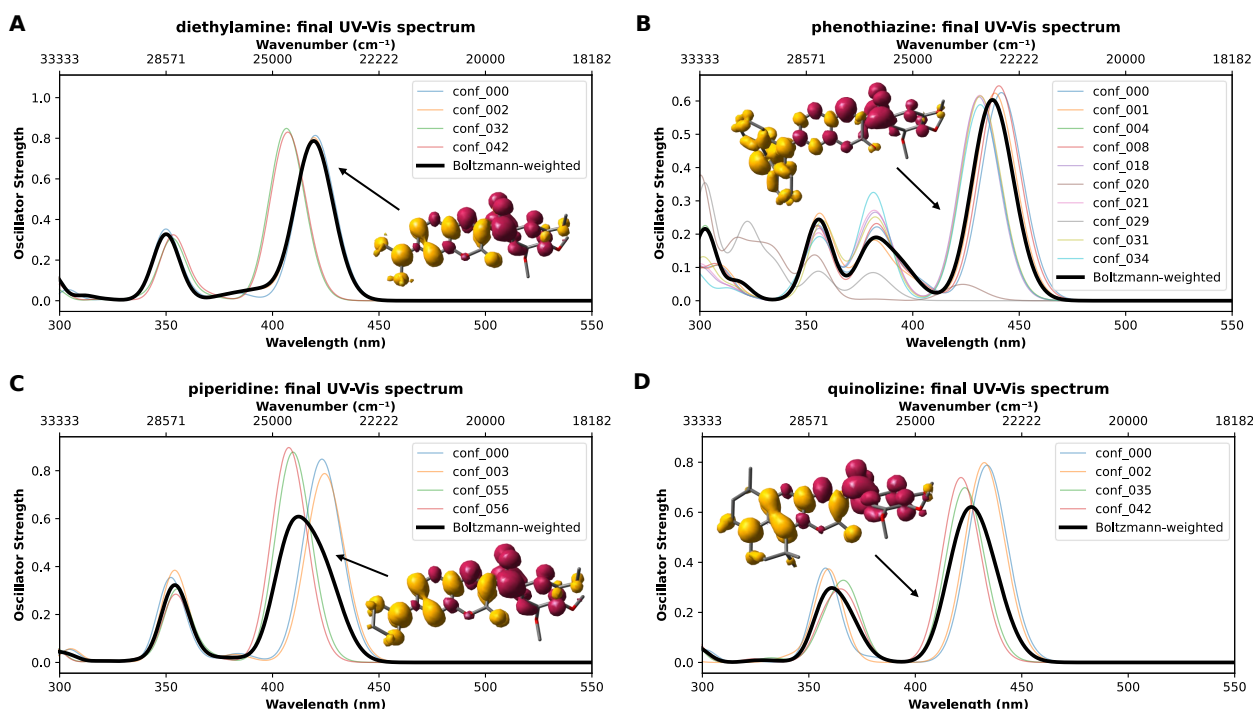


Figure 3: Calculated UV-Vis absorption spectra of the investigated ketocoumarins: (A) diethylamine, (B) phenothiazine, (C) piperidine, and (D) quinolizine. The thin curves show the spectra calculated for the individual DFT-optimized conformers. The thick black curves represent the final spectra obtained by weighting the individual conformer contributions according to their Boltzmann populations at 298.15 K. As an inset in each panel, we provide the electron density difference plot of the S_1 state, showing the character of the excitation. Yellow indicates loss of density and purple gain of density upon the excitation. An isovalue of 0.001 has been used.

For the diethylamine-substituted ketocoumarin, the final calculated absorption spectrum was dominated by *conf_002*. This conformer accounted for 95.18% of the calculated equilibrium population. The remaining three conformers each contributed less than 2% and therefore had only a minor influence on the final Boltzmann-weighted spectrum. Consequently, the position and shape of the principal absorption band were determined predominantly by the electronic transitions calculated for *conf_002*. The final spectrum exhibits a strong absorption band with a maximum near 420 nm and a weaker higher-energy band near 350 nm.

A similar conformational dependence was obtained for the phenothiazine-substituted ketocoumarin. Conformer *conf_004* accounted for 91.78% of the calculated equilibrium population and therefore provided the principal contribution to the final absorption spectrum. The conformers located approximately 2.4–2.6 kcal mol^{−1} above the minimum provided only small contributions to the combined spectral profile. The conformers located more than 5 kcal mol^{−1} above the minimum had negligible Boltzmann populations and therefore did not appreciably affect the final spectrum.

The piperidine-substituted ketocoumarin exhibited a different conformational contribution to its calculated absorption spectrum. All four optimized conformers had substantial populations ranging from 21.11% to 30.62%. No single conformer therefore dominated the conformational ensemble. The final spectrum represented a Boltzmann-weighted average over all four thermally accessible structures. Differences among the individual conformer spectra contributed to the overall shape and width of the calculated absorption bands.

For the quinolizine-substituted ketocoumarin, three conformers contributed substantially to the calculated spectrum. Conformers *conf_000*, *conf_035*, and *conf_042* accounted for 37.56%, 32.10%, and 29.86% of the equilibrium population, respectively. The final spectrum was therefore determined by the combined contributions of these three conformers. In contrast, *conf_002* contributed only 0.48% and con-

sequently had a negligible influence on the final absorption profile.

The calculated spectra demonstrate that the importance of conformational averaging differs considerably among the four ketocoumarins. For the diethylamine- and phenothiazine-substituted ketocoumarins, the final absorption profiles were governed predominantly by one conformer. For the piperidine- and quinoline-substituted ketocoumarins, several conformers contributed significantly and Boltzmann averaging was therefore essential for obtaining representative ensemble spectra. These results show that the calculated absorption profile depends not only on the chemical modification of the amine-containing fragment but also on the conformational distribution of the corresponding ketocoumarin.

The calculated absorption spectrum of the diethylamine-substituted ketocoumarin was compared with the experimental steady-state absorption spectrum of ketocoumarin 8a reported by Knorr, and co-workers [1]. The experimental spectrum was recorded in aerated methanol–water with a volume ratio of 70:30. In the published spectrum, the red curve represents steady-state absorption, whereas the blue curve represents fluorescence emission following excitation at 420 nm. Only the experimental absorption curve is therefore directly comparable with the calculated TD-DFT absorption spectrum.

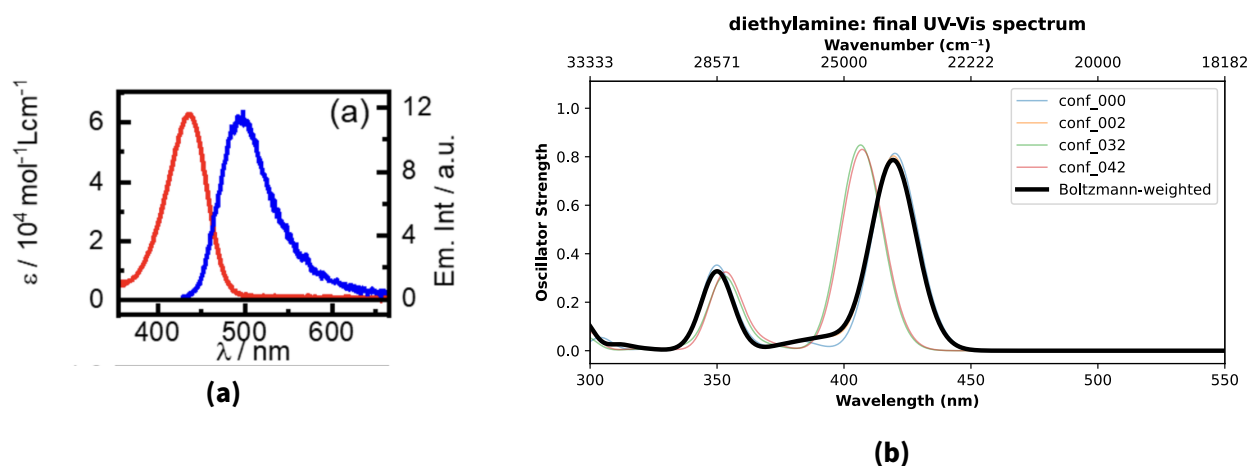


Figure 4: Comparison of the experimental and calculated absorption spectra of the diethylamine-substituted ketocoumarin. (a) Experimental steady-state absorption and emission spectra of ketocoumarin 8a in aerated methanol–water with a volume ratio of 70:30, adapted from Ref. [1]. The red curve represents absorption and the blue curve represents fluorescence emission. (b) Calculated Boltzmann-weighted TD-DFT absorption spectrum of the diethylamine-substituted ketocoumarin.

The experimental absorption maximum is located at 431 nm, whereas the principal maximum of the calculated Boltzmann-weighted spectrum is located at approximately 420 nm. The calculated maximum is therefore blue-shifted by approximately 10 nm relative to the experimental absorption maximum. This is a rather negligible shift and the calculation reproduces the position of the intense absorption band in the visible region reasonably well. The overall agreement indicates that the applied TD-DFT protocol provides a suitable qualitative and quantitative description of the absorption properties of the diethylamine-substituted ketocoumarin. We expect, therefore, an equally good performance for the remaining ketocoumarin molecules.

The remaining difference may originate from limitations of the applied density functional, the use of vertical excitation energies, the implicit treatment of the solvent, and the absence of explicit vibrational and solvent configurations. The calculated Gaussian broadening also produces a more idealized band shape than the experimentally measured spectrum.

4.3 Potential-Energy Curves along Important Torsions

Following the conformational analysis, the global minimum of each ketocoumarin (i.e. the conformer with the lowest free-energy) was selected as the starting structure for the torsional scans. The selected struc-

tures were `conf_002` for the diethylamine derivative, `conf_004` for the phenothiazine derivative, `conf_055` for the piperidine derivative, and `conf_000` for the quinolizine derivative. These conformers correspond to the lowest Gibbs free-energy structures obtained from the ground-state DFT optimizations and frequency calculations.

The main conformational degrees of freedom considered for the ketocoumarins are the rotations of the amine-containing substituent and that of the phenoxy group. Relaxed torsional scans were therefore performed along the corresponding dihedral coordinates. For the quinolizine derivative, only the phenoxy rotation was investigated because the amine-containing fragment is incorporated into a rigid fused-ring structure and cannot undergo an analogous torsional motion. For every investigated torsion, the scan was initiated from the dihedral angle of the selected optimized conformer. A total of 24 scan points were calculated using an angular increment of 7.5° . At each scan point, the selected dihedral angle was constrained, while the remaining molecular coordinates were optimized using ground-state DFT ('relaxed geometry scan'). The atom indices defining the torsional coordinates and the corresponding initial dihedral angles are summarized in Table 2.

Table 2: Torsional coordinates used for the relaxed ground-state DFT scans. Each scan consisted of 24 steps with an angular increment of 7.5° .

Ketocoumarin	Rotating group	Atom indices	Initial angle
Diethylamine	Amine	17–13–0–5	-175.60°
Diethylamine	Phenoxy	30–7–8–29	141.09°
Phenothiazine	Amine	41–60–0–5	-171.77°
Phenothiazine	Phenoxy	15–7–8–14	135.87°
Piperidine	Amine	41–55–0–5	-173.11°
Piperidine	Phenoxy	6–7–8–14	143.54°
Quinolizine	Phenoxy	0–1–4–10	143.57°

The relaxed ground-state DFT scans produced 24 optimized geometries for each investigated torsional coordinate. A separate TD-DFT calculation within the Tamm–Dancoff approximation was subsequently performed for every one of these geometries. Therefore, each torsional scan consisted of 24 relaxed ground-state DFT geometries followed by 24 corresponding TD-DFT calculations. For every scan geometry, the TD-DFT calculation provided the vertical excitation energies of the singlet and triplet excited states together with the oscillator strengths of the singlet transitions. The first three singlet states, S_1 , S_2 , and S_3 , and the first four triplet states, T_1 , T_2 , T_3 , and T_4 , were selected for the construction and analysis of the excited-state potential-energy curves. The calculated vertical excitation energies were combined with the corresponding ground-state energies at each scan point. This procedure yielded the potential-energy curves of S_1 , S_2 , S_3 , T_1 , T_2 , T_3 , and T_4 as functions of the scanned dihedral angle. The oscillator strengths of the singlet transitions were analysed simultaneously to monitor how the optical activity of the singlet excited states changed during rotation of the amine or phenoxy group.

4.3.1 The Rotation of the Amine Group

Relaxed scans were first performed along the amine torsional coordinate. In regions where an abrupt change between neighbouring optimized geometries was observed, short (reverse) relaxed scans were additionally carried out from a geometry on the other side of the discontinuity. This was due to somewhat big conformational changes during one step of the scan (e.g. reorientation of methyl groups) which resulted in these abrupt energy changes. The forward and reverse scan points were then combined to obtain a more continuous representation of the torsional potential-energy curves. TD-DFT calculations were performed for the selected relaxed geometries to follow the first three singlet and first four triplet excited states, while the oscillator strengths were used to monitor the optical activity of the singlet states. The results for the rotation of the amine group for ketocoumarins **A**, **B** and **C** are summarized in Figure 5

For the diethylamine derivative, the ground-state energy increases progressively during the first part of

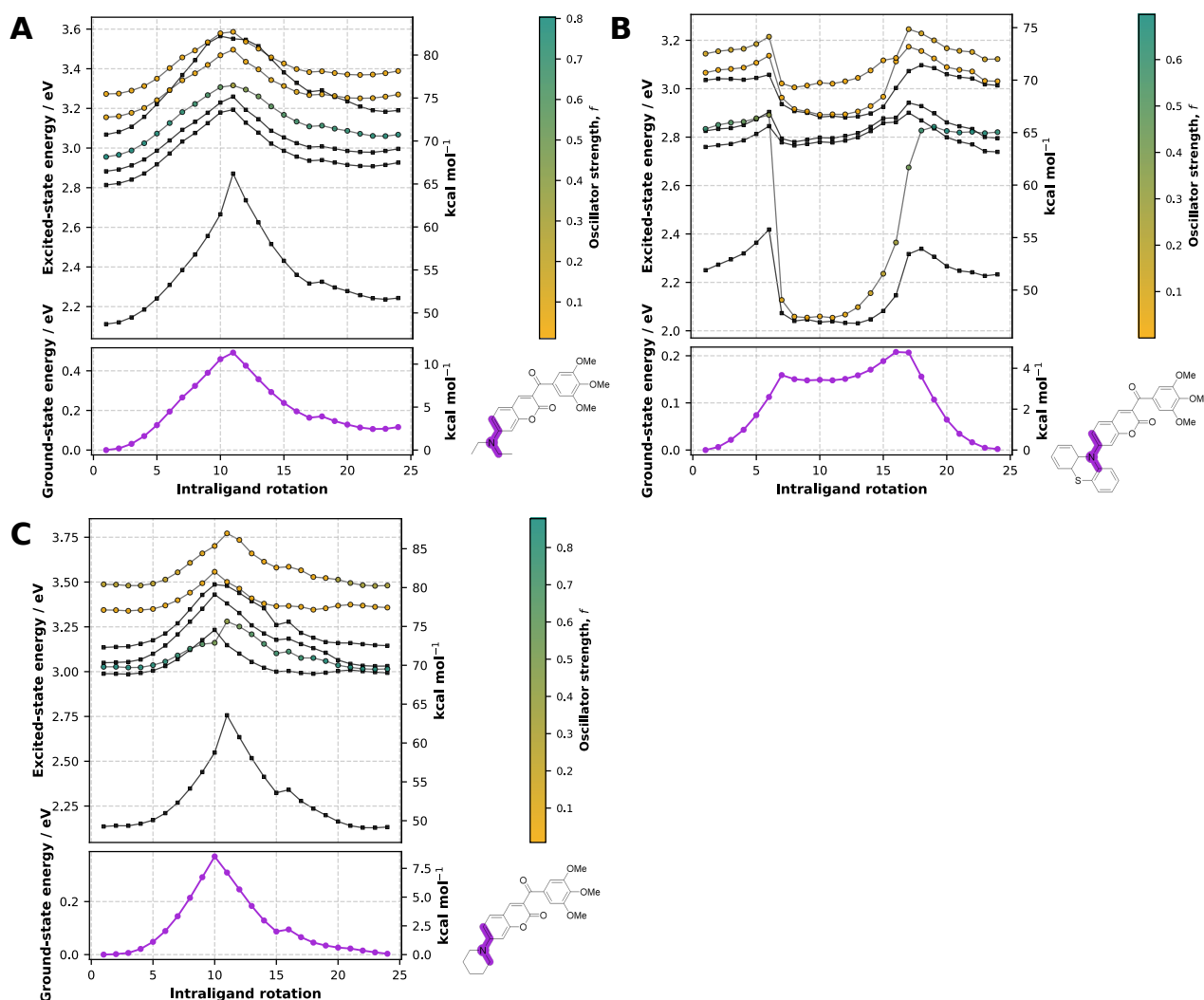


Figure 5: Potential-energy curves along the rotation of the amine group for (A) the diethylamine derivative, (B) the phenothiazine derivative, and (C) the piperidine derivative. The lower panels show the relative ground-state energies, while the upper panels show the corresponding excited-state energies. The oscillator strengths of the singlet excited states are represented by the colour scale.

the amine rotation and reaches its highest region around scan point 11, corresponding to a dihedral angle of approximately -90° , with an energy of approximately 0.5 eV (~ 11.5 kcal mol $^{-1}$) above the minimum. At this point, the molecule features a perpendicular orientation of the amine group. The energy subsequently decreases as the rotation proceeds and becomes considerably lower again in the second half of the scan. The energy subsequently decreases as the rotation proceeds and becomes considerably lower again in the second half of the scan. The excited states show a similar overall dependence on the amine torsion, with the excited-state energies increasing towards the central region of the scan and decreasing afterwards. For the three singlet states, however, the corresponding barriers are lower than that of the ground state. While the ground-state barrier is approximately 0.50 eV (~ 11.5 kcal mol $^{-1}$), the barriers along the S_1 , S_2 , and S_3 surfaces are approximately 0.36 eV (~ 8.3 kcal mol $^{-1}$), 0.34 eV (~ 7.8 kcal mol $^{-1}$), and 0.32 eV (~ 7.4 kcal mol $^{-1}$), respectively. This reduction indicates that rotation of the amine group along this torsional coordinate becomes energetically more accessible following excitation to the singlet excited states. Among the three singlet states, the lowest singlet state exhibits the largest oscillator strength over most of the investigated rotation, whereas the higher singlet states remain considerably weaker.

For the phenothiazine derivative, shown in panel B of Figure 5, the ground-state profile is comparatively

shallow, with the maximum of the barrier being approximately 0.21 eV (~ 4.8 kcal mol⁻¹). A more unusual behaviour is observed for the low-lying excited states, particularly S_1 and T_1 . Their potential-energy curves are divided into three distinct curved regions connected along the amine rotation, rather than following a single smooth parabolic profile. This behaviour suggests changes in the electronic character of the states along the torsional coordinate, consistent with state mixing between neighbouring electronic states. In the central region of the scan, the oscillator strength of the lowest singlet state is small, indicating a rather dark state in this part of the potential-energy surface. This low-energy region is centred around scan point 11, corresponding to a dihedral angle of approximately -90° , where the amine group is again perpendicular to the coumarin plane. In this perpendicular conformation, both S_1 and T_1 are strongly stabilized and approach one another energetically, being almost degenerate. This behaviour is particularly relevant for the function of the ketocoumarin as a photosensitizer, since the energetic proximity of the lowest singlet and triplet states may favour population transfer between them, although the potential-energy curves alone are not sufficient to establish the efficiency of intersystem crossing. Actually, the fact that the two states, around this part of the reaction coordinate, are nested and their orbital character (El Sayed's rule[22]) is the same (see Fig. 6) we can expect a rather inefficient population transfer. At the same time, the reduced energetic separation between the lowest excited state and the ground state, in this region, may also facilitate competing fluorescence or even non-radiative relaxation towards the ground state (the energy difference between the excited-state minimum and the ground state maximum corresponds to a wavelength of ca. 700 nm).

For the piperidine derivative, shown in panel C of Figure 5, the ground-state energy increases smoothly during the first part of the amine rotation and reaches a maximum around scan point 10 of approximately 0.32 eV (~ 7.4 kcal mol⁻¹) relative to the minimum. This scan point corresponds again to a perpendicular orientation. The energy then decreases again during the second half of the rotation, with only a small shoulder in the descending region, which is attributed to the patching between the forward and reverse scans. The excited-state potential-energy curves follow a similar overall trend, with their highest energies located around the central part of the torsional scan before decreasing towards the end of the rotation. The overall torsional behaviour of the piperidine derivative is similar to that of the diethylamine derivative. In both cases, the ground-state potential-energy surface exhibits a single broad barrier centred close to a perpendicular orientation of the amine torsion, and the excited-state energies increase towards this region before decreasing again. However, the ground-state barrier is clearly lower for the piperidine derivative, approximately 0.32 eV (~ 7.4 kcal mol⁻¹), compared with approximately 0.50 eV (~ 11.5 kcal mol⁻¹) for the diethylamine derivative. This indicates that rotation along the investigated amine torsional coordinate is energetically less demanding for the cyclic piperidine derivative. Since the principal structural difference between these two derivatives is the cyclic character of the aliphatic amine in the piperidine system, the comparison suggests that this structural constraint modifies the torsional energetics of the amine group.

In Figure 6, we present cartoons of the potential energy curves shown in Fig. 5, for the S_1 and T_1 states, along with electron density difference (EDD) maps for the excitations at some specific points of the potential curves (e.g. when a minimum or maximum is present).

4.3.2 The Rotation of the Phenoxy Group

Relaxed forward scans were first performed along the phenoxy torsional coordinate for all four ketocoumarins. In regions where abrupt changes between neighbouring optimized geometries were observed, short reverse relaxed scans were additionally performed from geometries located on the other side of the discontinuity. The forward and reverse scan points were subsequently combined to obtain a more continuous representation of the corresponding potential-energy curves. TD-DFT calculations were then performed for the selected relaxed geometries to follow the first three singlet and first four triplet excited states along the phenoxy rotation.

For the diethylamine derivative, shown in panel A of Figure 7, the ground-state profile exhibits two distinct barriers along the phenoxy rotation. The larger barrier occurs around scan point 8, corresponding to a dihedral angle of 86° , and reaches approximately 0.10 eV (~ 2.3 kcal mol⁻¹) relative to the minimum. A second, smaller barrier appears around scan point 19, corresponding to a dihedral angle of approximately -4° , and reaches approximately 0.07 eV (~ 1.6 kcal mol⁻¹). Between these two barriers, the ground-state

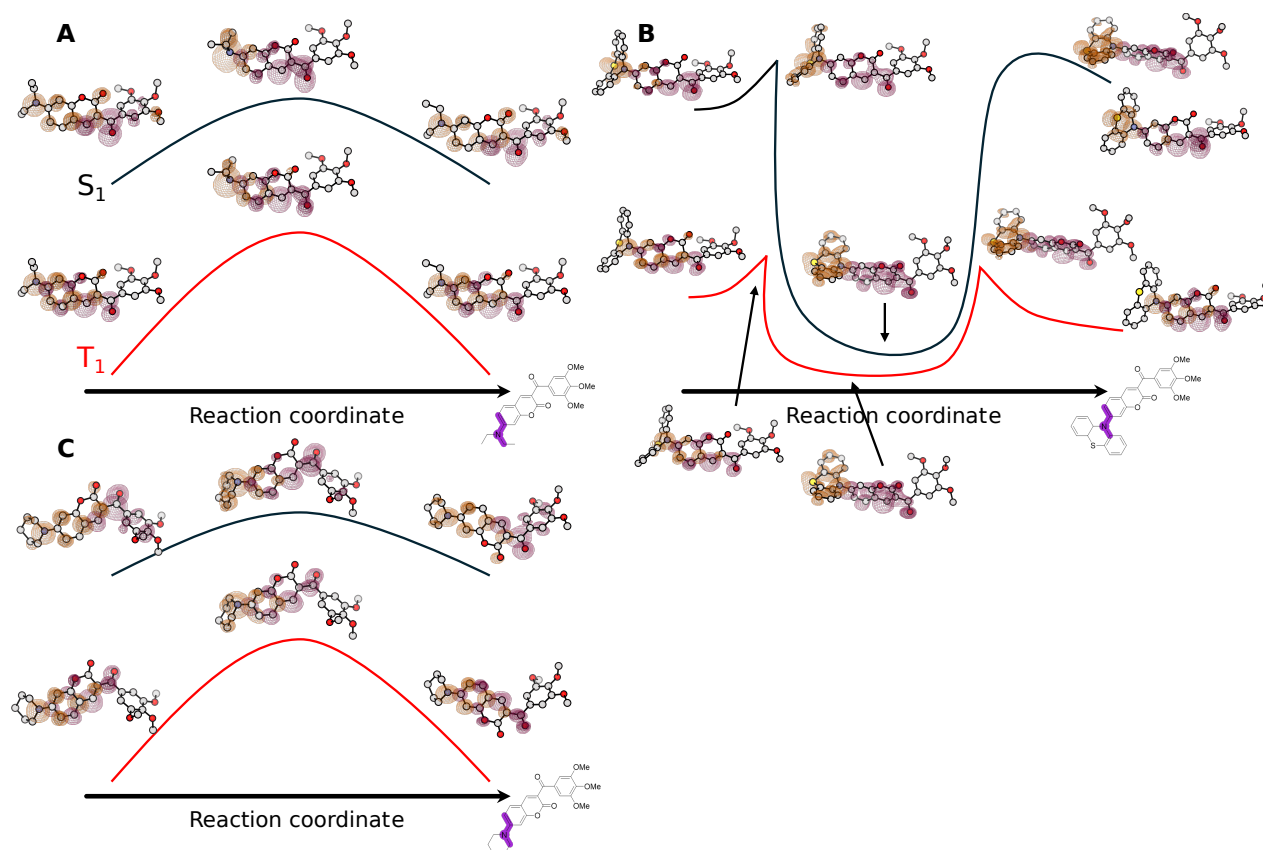


Figure 6: Schematically presented potential-energy curves of S_1 (black color) and T_1 (red color) along the rotation of the amine group for (A) the diethylamine derivative, (B) the phenothiazine derivative and (C) the piperidine derivative. Electron density difference (EDD) maps are shown for the respective excitations at specific points of the potential energy curve. Yellow indicates loss of density and purple gain of density upon the excitation. An isovalue of 0.0025 has been used.

energy decreases to a low-energy region around point 15, corresponding to phenoxy dihedral angles of approximately 31° . This region closely matches the phenoxy orientation of `conf_042` identified during the conformational sampling, for which the corresponding dihedral angle is approximately 48° . The decrease in energy in the middle of the scan can therefore be associated with the phenoxy group approaching one of the low-energy conformational orientations already identified during our conformational search. Towards the end of the scan, the phenoxy torsion also approaches the orientation represented by `conf_032`, which has a dihedral angle of approximately -47° . The excited-state potential-energy curves also depend strongly on the phenoxy rotation, particularly around the two barrier regions, where several excited states approach one another and changes in their energetic ordering are observed.

For the phenothiazine derivative, shown in panel B of Figure 7, the ground-state profile also exhibits two barriers along the phenoxy rotation. The first barrier occurs around scan point 7, corresponding to a dihedral angle of approximately 89° , and reaches approximately 0.08 eV (~ 1.8 kcal mol $^{-1}$) relative to the minimum. After decreasing to a minimum around scan points 13–14 (a dihedral of ca. 34°), the ground-state energy rises again to a slightly higher second barrier around scan point 18 of approximately 0.10 eV (~ 2.3 kcal mol $^{-1}$). The excited-state curves vary noticeably along the torsional coordinate, with the most pronounced changes occurring around the first barrier and smaller variations appearing in the region of the second barrier. Overall, the phenoxy rotation therefore produces two energetically distinct barrier regions in both the ground- and excited-state potential-energy profiles. The higher excited states show a behaviour similar to that observed for the diethylamine derivative, whereas the S_1 and T_2 states remain comparatively constant in energy around the first barrier. This reduced dependence on the phenoxy torsion suggests that

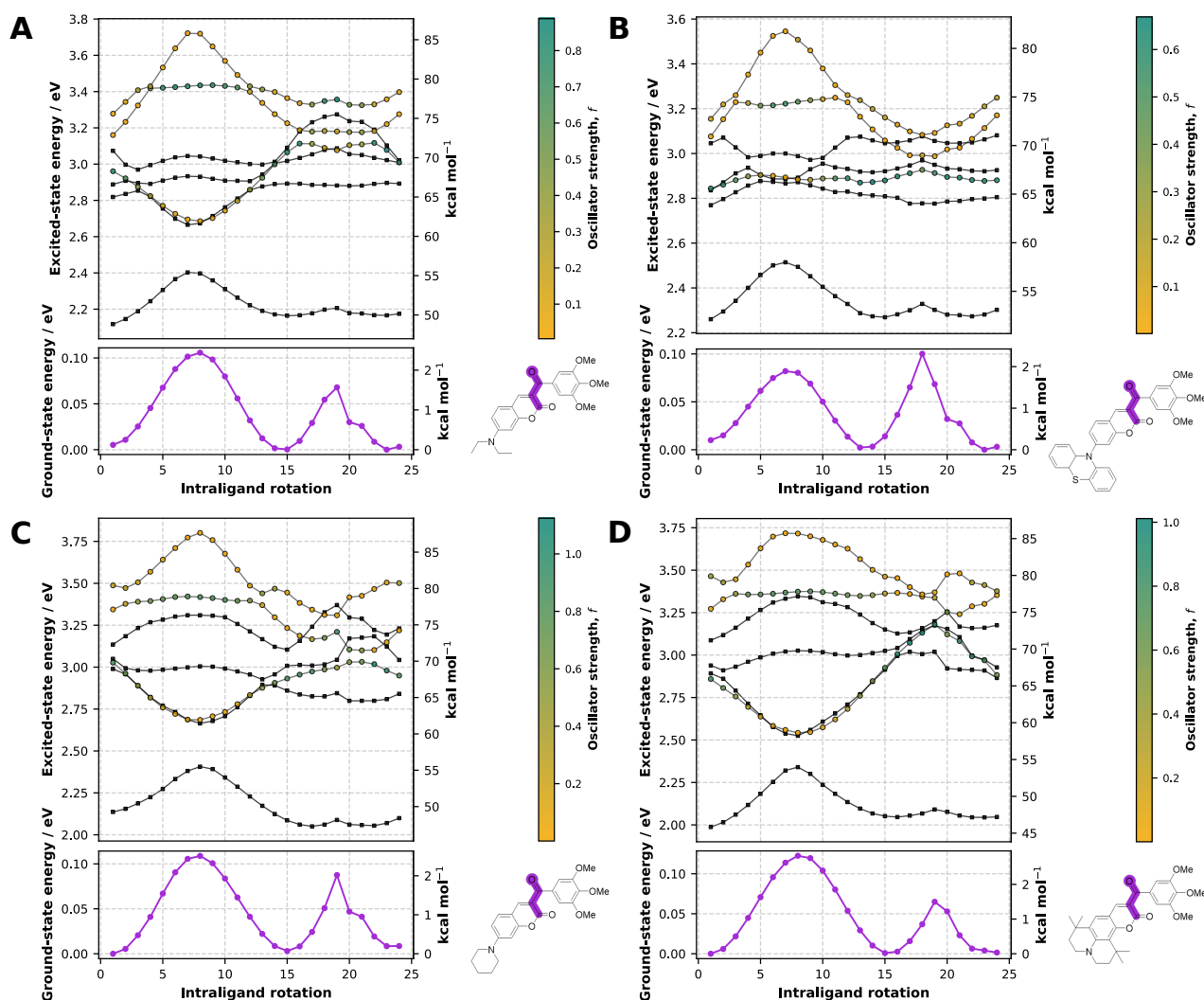


Figure 7: Potential-energy curves along the rotation of the phenoxy group for (A) the diethylamine derivative, (B) the phenothiazine derivative, (C) the piperidine derivative, and (D) the quinolizine derivative. The lower panels show the relative ground-state energies, while the upper panels show the corresponding excited-state energies. The colour scale represents the oscillator strengths of the singlet excited states.

the electronic character of these states is less strongly affected by the orientation of the phenoxy group than that of the higher excited states.

For the piperidine derivative, shown in panel C of Figure 7, the ground-state profile exhibits again two barriers along the phenoxy rotation. The first and slightly larger barrier occurs around scan point 8 and reaches approximately 0.11 eV (~ 2.5 kcal mol⁻¹) relative to the minimum. After decreasing to a second low-energy region around scan point 15, the energy rises to another barrier around scan point 19 of approximately 0.09 eV (~ 2.1 kcal mol⁻¹). The excited-state potential-energy curves vary smoothly over the first part of the scan but show more pronounced changes around scan points 18–21, where several states approach one another and their energetic ordering changes. Overall, the phenoxy rotation produces two relatively small ground-state barriers while having a more substantial influence on the arrangement of the excited states in the second half of the scan, in a very similar manner to **A**.

For the quinolizine derivative, shown in panel D of Figure 7, the ground-state profile exhibits once again two barriers along the phenoxy rotation, consistent with the other three KCs. The larger barrier occurs around scan point 8 and reaches approximately 0.12 eV (~ 2.8 kcal mol⁻¹) relative to the minimum. After decreasing to a minimum around scan point 15, the energy rises again to a smaller second barrier around

scan point 19 of approximately 0.065 eV (~ 1.5 kcal mol⁻¹). The excited-state potential-energy curves show a pronounced dependence on the phenoxy rotation, with several states changing substantially in energy over the scan. The most noticeable changes occur around scan points 18–21, where several excited states approach one another and their energetic ordering changes. Moreover, the avoided crossing around point 8 features a smaller energy difference compared to both **A** and **C**, indicating a smaller mixing (smaller coupling) between the states.

In Figure 8, we present cartoons of the potential energy curves shown in Fig. 7, for the S_1 and T_1 states, along with electron density difference (EDD) maps for the excitations at some specific points of the potential curves (e.g. when a minimum or maximum is present).

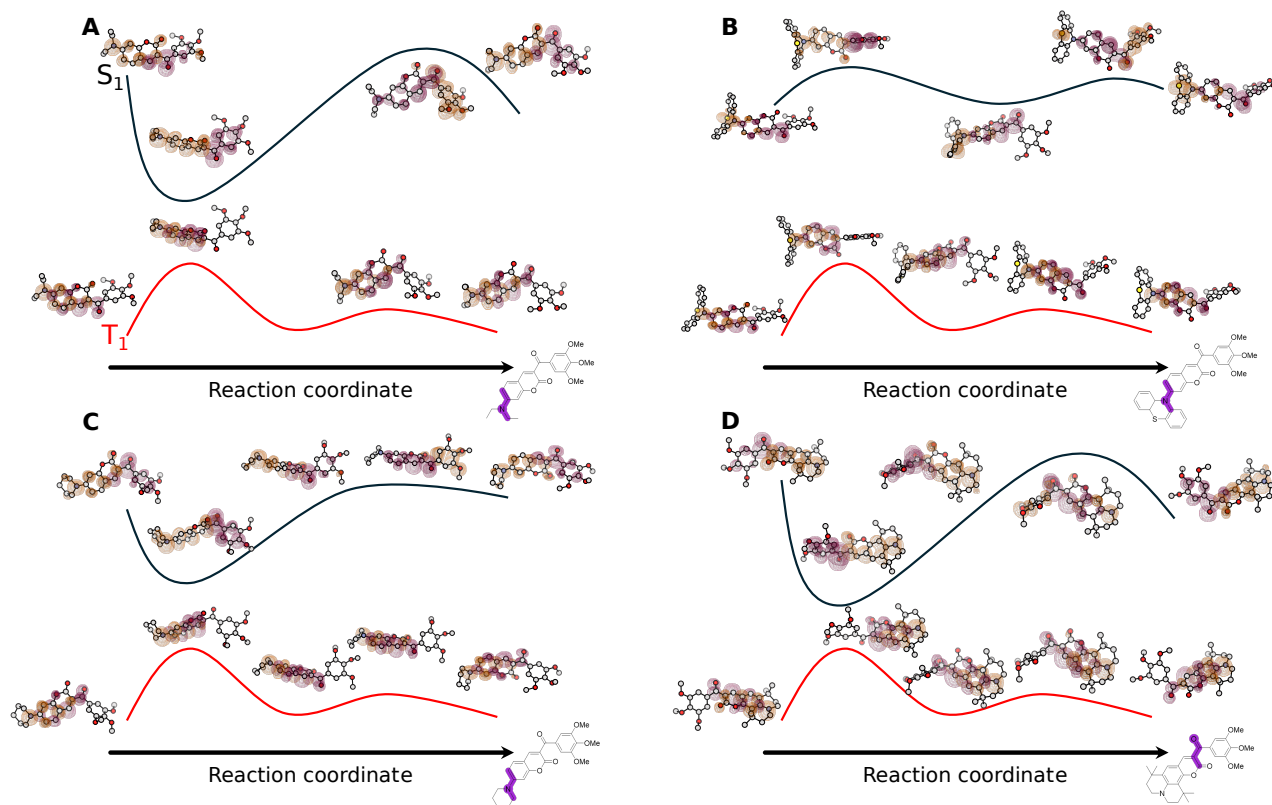


Figure 8: Schematically presented potential-energy curves of S_1 (black color) and T_1 (red color) along the rotation of the phenoxy group for (A) the diethylamine derivative, (B) the phenothiazine derivative, (C) the piperidine derivative and (D) the quinolizine derivative. Electron density difference (EDD) maps are shown for the respective excitations at specific points of the potential energy curve. Yellow indicates loss of density and purple gain of density upon the excitation. An isovalue of 0.0025 has been used.

5 Conclusions & Outlook

In this work, the conformational and electronic properties of four structurally modified ketocoumarin photosensitizers were investigated using conformational sampling, DFT optimizations, TD-DFT calculations, and relaxed torsional scans. The conformational analysis revealed clear differences between the investigated derivatives. The diethylamine and phenothiazine derivatives were strongly dominated by a single conformer, with calculated Boltzmann populations of 95.18% and 91.78%, respectively. In contrast, the piperidine derivative showed four similarly populated conformers, while three conformers contributed substantially to the equilibrium population of the quinolizine derivative. Consequently, conformational averaging was particularly important for describing the calculated absorption spectra of the piperidine and quinolizine derivatives.

The calculated UV-Vis spectrum of the diethylamine derivative reproduced the principal experimental absorption band reasonably well. The calculated maximum was located near 420 nm compared with the experimental value of 431 nm, corresponding to a relatively small blue shift of approximately 10 nm. The torsional scans further demonstrated that both the amine and phenoxy rotations can substantially modify the ground- and excited-state energy profiles. The amine rotation exhibited comparatively larger ground-state barriers, reaching approximately 11.5, 4.8, and 7.4 kcal mol⁻¹ for the diethylamine, phenothiazine, and piperidine derivatives, respectively. This indicates that in the ground state, the rotation of the amine group is rather restricted for all derivatives. In contrast, the phenoxy rotations showed considerably smaller barriers of approximately 1.5–2.8 kcal mol⁻¹, generally producing two barrier regions along the investigated coordinate and highlighting the relatively easier transformation between conformers. Particularly unusual behaviour was observed for the phenothiazine derivative during the amine rotation, where the S_1 and T_1 potential-energy curves suggest changes in electronic-state character and possible state mixing. The additional reverse scans helped to distinguish discontinuities caused by local conformational rearrangements and to construct more continuous and physically meaningful potential-energy curves.

These results provide a basis for a more detailed investigation of the excited-state dynamics of the ketocoumarins. Future work could extend the static potential-energy curves towards excited-state and non-adiabatic dynamics to investigate how the identified torsional motions influence intersystem crossing, formation of twisted charge-transfer states, and possible decomposition pathways at the ultrafast scale.

Future work could also extend the static potential-energy curves towards excited-state and non-adiabatic dynamics to investigate how the identified torsional motions influence intersystem crossing, formation of twisted charge-transfer states, and possible decomposition pathways at the ultrafast scale. Moreover, the calculated spectra could be further assessed through comparison with experimental spectra of the remaining ketocoumarin derivatives and through more detailed treatments of solvent and vibrational effects. Lastly, from a computational point of view, it is of interest to use a clustering algorithm after the results of conformational sampling to identify automatically the important conformers, without manual inspection.

A Brief Theoretical Background

The quantum-chemical methods used in this work are based on the electronic Schrödinger equation and its approximate treatment within density-functional theory. This appendix briefly summarizes the theoretical concepts that are directly related to the calculations performed for the ketocoumarin systems.

A.1 Electronic Schrödinger Equation and the Born–Oppenheimer Approximation

Within the Born–Oppenheimer approximation, the much heavier nuclei are treated as fixed while the electronic structure is solved for a given nuclear geometry. The electronic Schrödinger equation can therefore be written as

$$\hat{H}_{\text{el}}\Psi = E_{\text{el}}\Psi, \quad (6)$$

where $\hat{H}_{\text{el}}$ is the electronic Hamiltonian, Ψ is the many-electron wave function, and E_{el} is the corresponding electronic energy.

Repeating this calculation for different nuclear geometries defines a potential-energy surface on which molecular structures can be optimized. This concept is directly connected to the ground-state DFT geometry optimizations and relaxed torsional scans performed in this work. During the relaxed scans, one selected dihedral coordinate was constrained while the remaining nuclear coordinates were optimized, thereby following a one-dimensional section of the ground-state potential-energy surface.

A.2 Kohn–Sham Density-Functional Theory

Direct solution of the many-electron Schrödinger equation becomes computationally demanding because the wave function depends on the coordinates of all electrons. Density-functional theory instead describes the electronic system primarily through the electron density $\rho(\mathbf{r})$.

In Kohn–Sham DFT, the interacting many-electron system is represented by an auxiliary system of non-interacting electrons moving in an effective potential. The Kohn–Sham equations can be written as

$$\left[-\frac{1}{2}\nabla^2 + v_{\text{ext}}(\mathbf{r}) + v_{\text{H}}(\mathbf{r}) + v_{\text{xc}}(\mathbf{r}) \right] \phi_i(\mathbf{r}) = \varepsilon_i \phi_i(\mathbf{r}), \quad (7)$$

where v_{ext} describes the electron–nuclear interaction, v_{H} is the classical Coulomb contribution, and v_{xc} is the exchange–correlation potential. The orbitals ϕ_i are the Kohn–Sham molecular orbitals and ε_i are the corresponding orbital energies.

The total electronic energy can be expressed schematically as

$$E[\rho] = T_{\text{s}}[\rho] + E_{\text{ext}}[\rho] + E_{\text{H}}[\rho] + E_{\text{xc}}[\rho], \quad (8)$$

where T_{s} is the kinetic energy of the non-interacting Kohn–Sham system and E_{xc} contains the remaining exchange and correlation contributions. The exact form of E_{xc} is not known and must therefore be approximated.

A.3 Exchange–Correlation Functionals and TPSSh

Different levels of density-functional approximations can be distinguished according to the information used to construct the exchange–correlation functional. In the generalized-gradient approximation (GGA), the functional depends on both the local electron density and its spatial gradient,

$$E_{\text{xc}}^{\text{GGA}} = \int f(\rho(\mathbf{r}), \nabla\rho(\mathbf{r})) d\mathbf{r}. \quad (9)$$

Meta-GGA functionals include additional local information, commonly including the kinetic-energy density τ (that is $\nabla^2\rho$),

$$E_{xc}^{\text{meta-GGA}} = E_{xc}[\rho, \nabla\rho, \tau]. \quad (10)$$

Hybrid density functionals additionally incorporate a fraction of exact Hartree–Fock exchange. Their exchange–correlation energy can be represented schematically as

$$E_{xc}^{\text{hybrid}} = aE_x^{\text{HF}} + (1 - a)E_x^{\text{DFA}} + E_c^{\text{DFA}}, \quad (11)$$

where E_x^{HF} is the exact Hartree–Fock exchange contribution, E_x^{DFA} and E_c^{DFA} are the approximate density-functional exchange and correlation terms, and a determines the fraction of exact exchange.

The TPSSh functional used throughout this work is a hybrid meta-GGA functional based on the TPSS functional and contains 10% exact Hartree–Fock exchange [17]. It therefore combines the additional density information of a meta-GGA with a moderate contribution of exact exchange. TPSSh-D4 was used consistently for the DFT geometry optimizations, frequency calculations, relaxed torsional scans, and the subsequent TD-DFT calculations. The D4 dispersion correction was included to account for long-range dispersion interactions that are not described adequately by the underlying density functional alone.

A.4 Basis-Set Representation

The Kohn–Sham molecular orbitals are represented numerically as linear combinations of atom-centered basis functions. Within the linear-combination-of-atomic-orbitals approach, a molecular orbital can be written as

$$\phi_i(\mathbf{r}) = \sum_{\mu} C_{\mu i} \chi_{\mu}(\mathbf{r}), \quad (12)$$

where χ_{μ} denotes a basis function and $C_{\mu i}$ is the corresponding molecular-orbital coefficient.

The def2-TZVPP basis set was used in the present calculations [19, 20]. It provides a triple- ζ description of the valence orbitals together with additional polarization functions, allowing the electronic density to respond more flexibly to changes in molecular geometry and bonding.

The RIJCOSX approximation was used to reduce the computational cost associated with the Coulomb and exact-exchange contributions. The def2/J auxiliary basis set was employed for the corresponding density-fitting procedure. These approximations substantially improve computational efficiency while retaining the underlying TPSSh electronic-structure model.

A.5 Excited States and TD-DFT

Ground-state Kohn–Sham DFT provides the electronic energy and density of the ground state but does not directly provide the vertical electronic excitation energies required for an absorption spectrum. Time-dependent density-functional theory extends the DFT framework to the response of the electronic density to a time-dependent perturbation, such as an oscillating electromagnetic field.

The TD-DFT calculations performed in this work yielded vertical excitation energies and oscillator strengths for the electronic excited states. For a transition from the ground state 0 to an excited state I , the oscillator strength is related to the excitation energy and transition dipole moment and can be written schematically as

$$f_{0I} = \frac{2}{3} \omega_{0I} |\langle \Psi_0 | \hat{\mu} | \Psi_I \rangle|^2, \quad (13)$$

in atomic units, where ω_{0I} is the excitation energy and $\hat{\mu}$ is the electric-dipole operator. A larger oscillator strength therefore corresponds to a more strongly allowed electronic transition.

In the present work, the calculated excitation energies and oscillator strengths were used to construct the UV-Vis absorption spectra. For the torsional scans, the vertical excitation energies at each relaxed ground-state geometry were combined with the corresponding relative ground-state energy. This allowed the evolution of the singlet and triplet excited-state potential-energy curves to be followed as a function of the amine

and phenoxy torsional coordinates. Consequently, the electronic-structure theory summarized above provides the connection between the molecular geometries, ground-state energetics, absorption spectra, and excited-state potential-energy curves discussed throughout this report.

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